PERGAMON

Automatica 35 (1999) 1379–1390

www.elsevier.com/locate/automatica

Brief Paper

Adaptive control of robot manipulators with controller/update law modularity☆

M.S. de Queiroz[a,*], D.M. Dawson[b], M. Agarwal[b]

[a]*Department of Mechanical Engineering, Polytechnic University, Brooklyn, NY 11201, USA*
[b]*Department of Electrical and Computer Engineering, Clemson University, Clemson, SC 29634-0915, USA*

Received 25 March 1997; revised 3 July 1998; received in final form 4 January 1999

Abstract

In this paper, we present an adaptive link position tracking controller for robot manipulators which achieves controller/update law modularity. Specifically, the proposed torque input control law is designed to (i) achieve input-to-state stability with respect to the parameter estimation error, and (ii) guarantee asymptotic link position tracking and boundedness of all closed-loop signals for any adaptive update law that satisfies some generic properties. Simulation and experimental results are included to illustrate the advantages of the proposed adaptive control law. Specifically, the proposed controller with a least-squares estimator is compared to a well-known, gradient update-based, adaptive controller. The results indicate an improvement in the tracking performance with the proposed controller. © 1999 Elsevier Science Ltd. All rights reserved.

Keywords: Adaptive control; Robotic manipulators; Nonlinear control; Parameter estimation; Least-squares estimator

1 Introduction

Robot manipulators are nonlinear, multi-input/multi-output mechanical systems known to have a fairly well-defined dynamic structure; however, the physical parameters (e.g., payload masses, friction coefficients, etc.) used to complete the dynamic description are not often known precisely. This problem of parametric uncertainty has led many researchers to propose adaptive control laws that compensate for the unknown robot manipulator parameters while still maintaining accurate link position tracking. References to some representative work on adaptive robot control can be found in Queiroz et al. (1996).

Most of the above work employs a *Lyapunov-based* approach in the design of the adaptive controller (i.e., the controller and the adaptive update law are designed in conjunction via a single Lyapunov function). Unfortunately, this use of a single Lyapunov function often restricts the design of the adaptive update law. For example, many of the previous adaptive controllers utilize a standard gradient-type update law with the link position/velocity tracking error as the input. Since gradient-type update laws often exhibit slow parameter convergence in comparison with least-squares update laws, Lozano Leal and Canudas de Wit (1990) utilized the flexibility[1] provided by previous passivity-based adaptive control designs to construct a modified least-squares update law with the link position/velocity tracking error as the input. As time goes to infinity, the parameter estimates in this modified least-squares update law converge to the parameter estimates generated by a standard least-squares update law driven by the link position/velocity tracking error. A modified least-squares update law based on the link position/velocity tracking error was also proposed by Sadegh and Horowitz (1990) in the design of an exponentially stable DCAL-type controller. This exponential tracking result was predicated on the assumption that the *desired* regression matrix satisfies a *semi*-persistency of excitation condition;

* Corresponding author. Tel.: 001-718-260-3875; fax: 001-718-260-3532; e-mail: queiroz@poly.edu.

☆ This paper was not presented at any IFAC meeting. This paper was recommended for publication in revised form by Associate Editor R. Lozano under the direction of Editor C. C. Hang.

[1] The passivity-based adaptive controllers provide for some flexibility in the design of update law; however, the update law must be designed to satisfy a passive mapping condition.

0005-1098/99/$ - see front matter © 1999 Elsevier Science Ltd. All rights reserved
PII: S0005-1098(99)00048-5

furthermore, the modified least-squares update law requires the calculation of a matrix dependent on the excitation condition. Slotine and Li (1991) developed the *composite* adaptive controller by constructing the parameter update law as the composite sum of a least-squares update law driven by the *prediction*[2] error and a modified gradient update law driven by the link position/velocity tracking error (i.e., the gradient update law is modified because it uses the same time-varying adaptation gain as the least-squares update law). As compared to the standard persistency of excitation condition, the composite adaptive controller requires a weaker excitation condition (i.e., the infinite integral condition) for parameter convergence; however, this algorithm still utilizes a very specific choice for the adaptive update law. Tang and Arteaga (1994) developed an adaptive controller which included the standard gradient update law, the composite adaptation update law, and an averaging gradient update law as special cases.

Although mainly applied to linear systems, the *estimation-based* approach to adaptive control, as opposed to Lyapunov-based or passivity-based designs, allows for further flexibility in the construction of parameter update laws (e.g., prediction error-based gradient *or* normalized/unnormalized least-squares update laws can be designed). This flexibility is achieved due to the modular design of the controller and the update law. A modular estimation-based-type control approach for robot manipulators was first reported by Middleton and Goodwin (1988). Roughly speaking, in this work, the adaptive computed torque controller of Craig et al. (1986) was augmented with additional terms which allowed the closed-loop error system to be written as a stable, linear, strictly proper, transfer function with the link position tracking error as the output and a prediction error related term as the input. This input–output relationship facilitated the dual objectives of link position tracking and controller/update law modularity in the sense that any parameter update law could be used as long as its design ensured that: (i) the parameter estimates remain bounded, (ii) the prediction error is square integrable, and (iii) the estimated inertia matrix is positive-definite (PD) (i.e., a projection-type algorithm is required in the parameter update law). Recent work by Krstic and Kokotovic (1995) has successfully extended previous linear estimation-based techniques to a class of parametric-strict-feedback nonlinear systems by cleverly exploiting the *nonlinear damping* argument (Kokotovic, 1992). Unfortunately, the class of systems considered in Krstic and Kokotovic (1995) does not encompass the robot dynamics; moreover, due to the inertia matrix related coupling, it is not obvious how to extend the results given in Krstic and Kokotovic (1995) to achieve controller/update law modularity for robot manipulators. However, the work presented in Krstic and Kokotovic (1995) does provide some theoretical ammunition which can be used to attack the modularity problem for robot manipulators.

In this paper, we develop an adaptive link position tracking controller for robot manipulators which achieves controller/update law modularity. Specifically, we first design a torque input control law to achieve input-to-state stability with the parameter estimation error treated as a disturbance input to the link position tracking error dynamics; hence, the torque controller ensures closed-loop stability for any adaptive update law design which ensures that the parameter estimates remain bounded. We then illustrate how the design of the torque input control law yields global asymptotic link position tracking for any adaptive update law design which also satisfies the following sufficient conditions: (i) the prediction error is square integrable, and (ii) the estimated inertia matrix is PD. Hence, in contrast with the early work presented in Middleton and Goodwin (1988), closed-loop stability does not require the estimated inertia matrix to be PD; moreover, the proposed torque input control law does not require the online calculation of the inverse of the estimated inertia matrix as is required for the controller presented in Middleton and Goodwin (1988). To illustrate the advantages of the modular adaptive control design, the proposed control law using a standard least-squares estimator is compared to the well-known adaptive controller of Slotine and Li (1991) through dynamic simulations and experimental results.

The paper is organized as follows. Section 2 introduces the robot manipulator dynamic model along with some useful properties. In Section 3, we give the problem statement. Section 4 presents the input-to-state stability result while Section 5 presents the link position tracking result. Simulation and experimental results are presented in Section 6. Section 7 contains the conclusions.

2. Robot model

We will assume that the manipulator system is an n-link, serially connected, direct-drive, rigid-link, revolute robot with the following dynamic model (Lewis et al., 1993):

$$M(q)\ddot{q} + V_m(q, \dot{q})\dot{q} + G(q) + F_d\dot{q} = \tau, \tag{1}$$

where $q(t), \dot{q}(t), \ddot{q}(t) \in \Re^n$ denote the link position, velocity, and acceleration vectors, respectively, $M(q) \in \Re^{n\times n}$ is the inertia matrix, $V_m(q, \dot{q}) \in \Re^{n\times n}$ is the centripetal-Coriolis matrix, $G(q) \in \Re^n$ represents the gravity vector,

[2] In the composite adaptive controller, the prediction error is defined as the difference between an estimated, filtered version of the robot dynamics and a filtered version of the input torque.

$F_d \in \Re^{n\times n}$ is a PD, diagonal matrix representing the viscous friction coefficients, and $\tau \in \Re^n$ is the torque input vector.

The robot dynamic equation of (1) has the following properties (Lewis et al., 1993) that will be used later in the controller-parameter update law development and analysis.

Property 1 (*Boundedness of the inertia matrix*). The inertia matrix $M(q)$ is symmetric, PD, and satisfies the following inequalities:

$$m_1\|y\|^2 \le y^{\mathrm{T}}M(q)y \le m_2\|y\|^2 \quad \forall y \in \Re^n, \tag{2}$$

where m_1, m_2 are known positive constants, and $\|\cdot\|$ is the standard Euclidean norm.

Property 2 (*Skew symmetry*). For a proper definition of the centripetal-Coriolis matrix, the following skew symmetric relationship is satisfied:

$$y^{\mathrm{T}}(\tfrac{1}{2}\dot{M}(q) - V_m(q,\dot{q}))y = 0 \quad \forall y \in \Re^n, \tag{3}$$

where $\dot{M}(q)$ is the time derivative of the inertia matrix.

Property 3 (*Linearity in the parameters*). The dynamic equation of (1) is linear parameterizable in the following manner:

$$M(q)\ddot{q} + V_m(q,\dot{q})\dot{q} + G(q) + F_d\dot{q} = Y(q,\dot{q},\ddot{q})\theta, \tag{4}$$

where $\theta \in \Re^p$ contains the unknown, constant system parameters, and $Y(q,\dot{q},\ddot{q}) \in \Re^{n\times p}$, called the regression matrix, contains known functions of the link position, velocity, and acceleration.

3. Problem statement

The problem is to design an adaptive link position tracking control system for the robot dynamics of (1) such that a modularity is achieved in the design of the controller and the parameter update law. To accomplish this purpose, we first define the link position tracking error $e(t) \in \Re^n$ and the corresponding filtered tracking error $r(t) \in \Re^n$ (Slotine and Li, 1991) as

$$e = q_{\mathrm{d}} - q, \qquad r = \dot{e} + \alpha e, \tag{5}$$

where $q_{\mathrm{d}}(t) \in \Re^n$ denotes the desired link position trajectory (we assume $q_{\mathrm{d}}(t)$ and its first and second time derivatives are bounded), and $\alpha \in \Re^{n\times n}$ is a PD, diagonal, control gain matrix. Since the control objective is to be met under the constraint of parametric uncertainty, the controller will contain a parameter update law to estimate the unknown parameters; hence, the *parameter* error will be denoted by

$$\tilde{\theta} = \theta - \hat{\theta}, \tag{6}$$

where $\hat{\theta}(t) \in \Re^p$ is a dynamic estimate of θ, previously defined in (4). Furthermore, we define the *filtered* regression matrix (Middleton and Goodwin, 1988) $Y_f(q,\dot{q}) \in \Re^{n\times p}$ as

$$\begin{aligned} &\dot{Y}_f(q,\dot{q}) + \beta Y_f(q,\dot{q}) = \beta Y(q,\dot{q},\ddot{q}),\\ &Y_f(q(0),\dot{q}(0)) = 0, \end{aligned} \tag{7}$$

where $Y(\cdot)$ was defined in (4) (see Lewis et al. (1993) for details on how to compute $Y_f(\cdot)$ without the need for acceleration measurements), β is a positive scalar constant, and the measurable *prediction* error $\varepsilon(t) \in \Re^n$ is defined as

$$\varepsilon = Y_f\tilde{\theta} = Y_f\theta - Y_f\hat{\theta} = \tau_f - Y_f\hat{\theta}, \tag{8}$$

where the filtered torque $\tau_f(t) \in \Re^n$ is given by

$$\dot{\tau}_f + \beta\tau_f = \beta\tau, \qquad \tau_f(0) = 0 \tag{9}$$

with $\tau(t)$ being defined in (1).

4. Input-to-state stability result

One of the main control objectives is to achieve input-to-state stability (ISS) with respect to the *disturbance input* $\tilde{\theta}(t)$ (Krstic and Kokotovic, 1995). To formulate the torque input control law, we first rewrite the robot dynamics of (1) in terms of the filtered tracking error definition, given in (5), as follows:

$$M(q)\dot{r} = Y_s(q,\dot{q},t)\theta - \tau - V_m(q,\dot{q})r, \tag{10}$$

where the linear parameterization $Y_s(\cdot)\theta$ is defined as

$$\begin{aligned} Y_s(q,\dot{q},t)\theta = {} & M(q)(\ddot{q}_{\mathrm{d}} + \alpha\dot{e}) + V_m(q,\dot{q})(\dot{q}_{\mathrm{d}} + \alpha e)\\ & + G(q) + F_d\dot{q} \end{aligned} \tag{11}$$

with $Y_s(\cdot) \in \Re^{n\times p}$ being a known regression matrix. Based on the stated control objective, the form of (10), and the subsequent analysis, we propose the following control law:

$$\begin{aligned} \tau = {} & Kr + Y_s\hat{\theta} + \left[\frac{1}{\beta}Y_f\dot{\hat{\theta}}\right] + [(\dot{\hat{M}} - \hat{V}_m)r] + k_n\|Y_s\|^2 r\\ & + k_n\left\|\frac{1}{\beta}Y_f\dot{\hat{\theta}}\right\|^2 r + k_n\|(\dot{\hat{M}} - \hat{V}_m)r\|^2 r, \end{aligned} \tag{12}$$

where $K \in \Re^{n\times n}$ is a PD, diagonal, control gain matrix, k_n is a positive scalar, nonlinear damping gain (Kokotovic, 1992) $\hat{M}(q)$, $\hat{V}_m(q,\dot{q})$ represent the estimates of the inertia and centripetal-Coriolis matrices, and $Y_f(\cdot)$ was defined in (7). By substituting (12) into (10), we can form the closed-loop filtered tracking error

system as follows:

$$\begin{aligned} M(q)\dot{r} = & -Kr + Y_s\tilde{\theta} - k_n\|Y_s\|^2 r \\ & - \frac{1}{\beta} Y_f\dot{\hat{\theta}} - k_n\left\|\frac{1}{\beta} Y_f\dot{\hat{\theta}}^2\right\| r \\ & - (\dot{\hat{M}} - \hat{V}_m)r - k_n\|(\dot{\hat{M}} - \hat{V}_m)r\|^2 r \\ & - V_m(q, \dot{q})\, r. \end{aligned} \tag{13}$$

Remark 1. With respect to the control law of (12), we first note that the bracketed terms have been introduced to cancel similar terms in the subsequent stability analysis. The norm squared terms of (12) are nonlinear damping functions injected to accomplish the ISS property of the closed-loop system with respect to $\tilde{\theta}(t)$.

We now state a theorem regarding the ISS property of (13) with respect to $\tilde{\theta}(t)$.

Theorem 1. *Given the closed-loop error system of* (13), *if* $\tilde{\theta}(t) \in \mathscr{L}_\infty[0, t_f)$ *then* $r(t) \in \mathscr{L}_\infty[0, t_f)$ *where* t_f *denotes the final time.*

Proof. We start by defining the nonnegative function

$$V_1 = \tfrac{1}{2} r^{\mathrm{T}} M(q) r. \tag{14}$$

After taking the time derivative of (14) along (13), we obtain the following upper bound on $\dot{V}_1$:

$$\begin{aligned} \dot{V}_1 \le & -\lambda_{\min}\{K\}\|r\|^2 + [\|Y_s\|\|r\|\|\tilde{\theta}\| - k_n\|Y_s\|^2\|r\|^2] \\ & + \left[\left\|\frac{1}{\beta} Y_f\dot{\hat{\theta}}\right\|\|r\| - k_n\left\|\frac{1}{\beta} Y_f\dot{\hat{\theta}}\right\|^2\|r\|^2\right] \\ & + [\|(\dot{\hat{M}} - \hat{V}_m)r\|\|r\| - k_n\|(\dot{\hat{M}} - \hat{V}_m)r\|^2\|r\|^2], \end{aligned} \tag{15}$$

where (3) has been utilized, and $\lambda_{min}\{\cdot\}$ denotes the minimum eigenvalue of a matrix. After applying the nonlinear damping argument (Kokotovic, 1992) to the bracketed terms of (15), we have

$$\dot{V}_1 \le -\lambda_{\min}\{K\}\|r\|^2 + \frac{1}{k_n}\|\tilde{\theta}\|^2 + \frac{2}{k_n} \tag{16}$$

which can be further upper bound by the use of (2) and (14) as

$$\dot{V}_1(t) \le -\lambda V_1(t) + \frac{1}{k_n}\|\tilde{\theta}(t)\|^2 + \frac{2}{k_n}, \tag{17}$$

where $\lambda = 2\lambda_{\min}\{K\}/m_2$. After solving the differential inequality of (17), we have

$$\begin{aligned} V_1(t) & \le V_1(0)\,\mathrm{e}^{-\lambda t} + \mathrm{e}^{-\lambda t}\int_0^t \mathrm{e}^{\lambda\xi}\left(\frac{1}{k_n}\|\tilde{\theta}(\xi)\|^2 + \frac{2}{k_n}\right)\mathrm{d}\xi \\ & \le V_1(0)\,\mathrm{e}^{-\lambda t} + \frac{1}{\lambda k_n}(\|\tilde{\theta}(t)\|_\infty^2 + 2), \end{aligned} \tag{18}$$

where $\|\cdot\|_\infty$ denotes the $\mathscr{L}_\infty$ norm of a signal. Hence, from (18), (2), and (14), we have

$$\begin{aligned} \|r(t)\| \le & \sqrt{\frac{m_2}{m_1}}\,\|r(0)\|\mathrm{e}^{-\lambda/2t} \\ & + \sqrt{\frac{2}{m_1\lambda k_n}\left(\|\tilde{\theta}(t)\|_\infty^2 + 2\right)} < \infty \end{aligned} \tag{19}$$

which proves the theorem. □

Remark 2. Since the link position tracking error is related to the filtered tracking error according to a stable linear differential equation given by (5), it is easy to show (Desoer and Vidyasagar, 1975) that $e(t), \dot{e}(t) \in \mathscr{L}_\infty[0, t_f)$; hence, $q(t), \dot{q}(t) \in \mathscr{L}_\infty[0, t_f)$ due to boundedness of the desired link trajectory.

5. Link position tracking result

The control law (12) guarantees the boundedness of the system states independent of the adaptation algorithm (i.e., it permits flexibility in the design of the parameter update law as long as the boundedness condition on $\tilde{\theta}(t)$ is satisfied). We now turn our attention to developing specific technical conditions on the update law which leads to boundedness of all closed-loop signals and link position tracking. To this end, we state the following theorem.

Theorem 2. *Given the control law* (12) *and any update law, denoted by* $\dot{\hat{\theta}}(q, \dot{q}, \varepsilon)$, *that ensures* $\hat{\theta}(t) \in \mathscr{L}_\infty[0, t_f)$, *we can state that*: (i) *all signals are bounded during closed-loop operation on* $[0, \infty)$, *and* (ii) *if the update law also ensures that* $\varepsilon(t) \in \mathscr{L}_2[0, t_f)$ *then* $\lim_{t\to\infty} e(t), \dot{e}(t) = 0$ *provided* $\hat{M}(q)$ *remains PD and* K *of* (12) *is selected as* $K = kI_n$ *where* k *is a positive scalar, and* I_n *is the* $n \times n$ *identity matrix.*

Proof of (i). Since $\hat{\theta}(t) \in \mathscr{L}_\infty[0, t_f)$, we know from Theorem 1 and Remark 2 that $r(t), q(t), \dot{q}(t) \in \mathscr{L}_\infty[0, t_f)$; hence, $Y_f(q, \dot{q}) \in \mathscr{L}_\infty[0, t_f)$. Since $\hat{\theta}(t), Y_f(q, \dot{q}) \in \mathscr{L}_\infty[0, t_f)$, we can state from (8) that $\varepsilon(t) \in \mathscr{L}_\infty[0, t_f)$. Since $\dot{\hat{\theta}}(t)$ is a function of $q(t)$, $\dot{q}(t)$, and $\varepsilon(t)$, we know that $\dot{\hat{\theta}}(t) \in \mathscr{L}_\infty[0, t_f)$. From the previous boundedness statements, we can see from (13) and (12) that $\dot{r}(t)$, $\tau(t) \in \mathscr{L}_\infty[0, t_f)$, and therefore, $\ddot{q}(t) \in \mathscr{L}_\infty[0, t_f)$ from (1) and (2). Thus, we have shown that all closed-loop signals are bounded on $[0, t_f)$. As in Krstic and Kokotovic (1995), the bounds are dependent only on the initial conditions, control gains, and the desired motion trajectory (i.e., not dependent on t_f); hence, this independence on t_f proves that $t_f = \infty$. □

Proof of (ii). We start by noting from (4) and (11) that

$$Y(q, \dot{q}, \ddot{q})\tilde{\theta} = Y_s(q, \dot{q}, t)\tilde{\theta} - \tilde{M}(q)\dot{r} - \tilde{V}_m(q, \dot{q})r. \tag{20}$$

where

$$\tilde{M}(q) = M(q) - \hat{M}(q)$$

and

$$\tilde{V}_m(q, \dot{q}) = V_m(q, \dot{q}) - \hat{V}_m(q, \dot{q}).$$

With the intent of writing the term $Y_s\tilde{\theta}$ in terms of $\varepsilon(t)$, we can use (8), the time derivative of (6), and the relationships given by (7) and (20) to show that

$$\begin{aligned}\frac{1}{\beta}\dot{\varepsilon} + \varepsilon &= \frac{1}{\beta}\frac{\mathrm{d}}{\mathrm{d}t}(Y_f\tilde{\theta}) + Y_f\tilde{\theta} \\ &= \frac{1}{\beta}\dot{Y}_f\tilde{\theta} + Y_f\tilde{\theta} - \frac{1}{\beta}Y_f\dot{\hat{\theta}} \\ &= Y\tilde{\theta} - \frac{1}{\beta}\dot{Y}_f\dot{\hat{\theta}} \\ &= Y_s\tilde{\theta} - \tilde{M}(q)\dot{r} - \tilde{V}_m(q, \dot{q})r - \frac{1}{\beta}Y_f\dot{\hat{\theta}}. \end{aligned} \tag{21}$$

We can now utilize the last line in (21) to obtain the following expression for $Y_s\tilde{\theta}$

$$Y_s\tilde{\theta} = \frac{1}{\beta}\dot{\varepsilon} + \varepsilon + \tilde{M}(q)\,\dot{r} + \tilde{V}_m(q, \dot{q})r + \frac{1}{\beta}Y_f\,\dot{\hat{\theta}}. \tag{22}$$

After substituting the right-hand side of (22) into (13), we have

$$\hat{M}(q)\dot{r} = -A(q, \dot{q}, \hat{\theta}, \dot{\hat{\theta}}, t)r + \frac{1}{\beta}\dot{\varepsilon} + \varepsilon - \dot{\hat{M}}(q)\,r, \tag{23}$$

where the *positive* scalar function $A(\cdot)$ is defined as

$$\begin{aligned}A(q, \dot{q}, \hat{\theta}, \dot{\hat{\theta}}, t) &= k + k_n\|Y_s\|^2 + k_n\left\|\frac{1}{\beta}Y_f\,\dot{\hat{\theta}}\right\|^2 \\ &\quad + k_n\|(\dot{\hat{M}} - \hat{V}_m)r\|^2. \end{aligned} \tag{24}$$

We now define the following variable transformation:

$$x = \hat{M}(q)r - \frac{1}{\beta}\varepsilon. \tag{25}$$

After taking the time derivative of (25) and substituting (23) into the resulting expression, we obtain

$$\dot{x} = -A(\cdot)r + \varepsilon. \tag{26}$$

By utilizing (25), the above expression for $\dot{x}$ can be rewritten as

$$\dot{x} = -A\hat{M}^{-1}x + B\varepsilon, \tag{27}$$

where the matrix $B(\cdot) \in \Re^{n\times n}$ is defined as

$$B(q, \dot{q}, \hat{\theta}, .\hat{\theta}, t) = \left(I_n - \frac{1}{\beta}A\hat{M}^{-1}\right). \tag{28}$$

After differentiating $\frac{1}{2}x^{\mathrm{T}}x$ along (27), we obtain[3]

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(\tfrac{1}{2}x^{\mathrm{T}}x) &= -Ax^{\mathrm{T}}\hat{M}^{-1}x + x^{\mathrm{T}}B\varepsilon \\ &\le -A\lambda_{\min}\{\hat{M}^{-1}\}\|x\|^2 + \|B\|_\infty\|\varepsilon\|\|x\|. \end{aligned} \tag{29}$$

We now let

$$y = \sqrt{\tfrac{1}{2}x^{\mathrm{T}}x} \tag{30}$$

and rewrite (29) as

$$\frac{\mathrm{d}}{\mathrm{d}t}(y^2) = 2y\dot{y} \le -A\lambda_{\min}\{\hat{M}^{-1}\}y^2 + \sqrt{2}\|B\|_\infty\|\varepsilon\|y. \tag{31}$$

Since $y(t) \ge 0$, we can use (31) to produce the following upper bound on $\dot{y}(t)$:

$$\dot{y} \le -\zeta_1 y + \zeta_2\|\varepsilon\|, \tag{32}$$

where the *positive* scalar constants[4] ζ_1 and ζ_2 are defined as

$$\zeta_1 = \tfrac{1}{2}\inf_t\{A\}\lambda_{\min}\{\hat{M}^{-1}\} \quad \text{and} \quad \zeta_2 = \frac{\sqrt{2}}{2}\|B\|_\infty. \tag{33}$$

The solution of the differential inequality of (32) is given by

$$y(t) \le y(0)\mathrm{e}^{-\zeta_1 t} + \zeta_2\int_0^t \mathrm{e}^{-\zeta_1(t-\xi)}\|\varepsilon(\xi)\|\,\mathrm{d}\xi. \tag{34}$$

Utilizing Holder's inequality (Vidyasagar, 1978), (34) can be rewritten as follows:

$$\begin{aligned}|y(t)| &\le |y(0)|\mathrm{e}^{-\zeta_1 t} + \zeta_2\sqrt{\textstyle\int_0^t \mathrm{e}^{-\zeta_1(t-\xi)}\,\mathrm{d}\xi} \\ &\qquad \sqrt{\textstyle\int_0^t \mathrm{e}^{-\zeta_1(t-\xi)}\|\varepsilon(\xi)\|^2\,\mathrm{d}\xi} \\ &\le |y(0)|\mathrm{e}^{-\zeta_1 t} + \frac{\zeta_2}{\sqrt{\zeta_1}}\sqrt{\textstyle\int_0^t \mathrm{e}^{-\zeta_1(t-\xi)}\|\varepsilon(\xi)\|^2\,\mathrm{d}\xi}. \end{aligned} \tag{35}$$

After squaring (35) and integrating the resulting expression, we obtain

$$\begin{aligned}\int_0^t |y(\sigma)|^2\,\mathrm{d}\sigma &\le \frac{|y(0)|^2}{\zeta_1} \\ &\quad + \frac{2\zeta_2^2}{\zeta_1}\int_0^t\left(\int_0^\sigma \mathrm{e}^{-\zeta_1(\sigma-\xi)}\|\varepsilon(\xi)\|^2\,\mathrm{d}\xi\right)\mathrm{d}\sigma. \end{aligned} \tag{36}$$

[3] Note that if $\hat{M}(q)$ is PD, we know $\hat{M}^{-1}(q)$ is also PD.

[4] Since we know all signals are bounded from part (i) of Theorem 2, the constants ζ_1 and ζ_2 exist.

After reversing the order of integration in (36), we obtain

$$\int_0^t |y(\sigma)|^2 \,\mathrm{d}\sigma \le \frac{|y(0)|^2}{\zeta_1} + \frac{2\zeta_2^2}{\zeta_1}\int_0^t \mathrm{e}^{\zeta_1\xi}\|\varepsilon(\xi)\|^2\left(\int_\xi^t \mathrm{e}^{-\zeta_1\sigma}\,\mathrm{d}\sigma\right)\mathrm{d}\xi$$
$$\le \frac{|y(0)|^2}{\zeta_1} + \frac{2\zeta_2^2}{\zeta_1}\int_0^t \|\varepsilon(\xi)\|^2\frac{1}{\zeta_1}\,\mathrm{d}\xi. \tag{37}$$

After utilizing the fact that $\varepsilon(t)\in\mathscr{L}_2$, we have from (37) that

$$\|y\|_2 \le \frac{|y(0)|^2}{\zeta_1} + \frac{2\zeta_2^2}{\zeta_1^2}\|\varepsilon\|_2 < \infty, \tag{38}$$

where $\|\cdot\|_2$ denotes the $\mathscr{L}_2$ norm of a signal. The result given by (38) illustrates that $y(t)\in\mathscr{L}_2$. From the definition of $y(t)$ given in (30), we can conclude $x(t)\in\mathscr{L}_2$. Since $\hat{M}(q)$ is assumed to be PD, we can now use the variable transformation of (25) to show that $r(t)\in\mathscr{L}_2$. Since $r(t)\in\mathscr{L}_\infty\cap\mathscr{L}_2$ and $\dot{r}(t)\in\mathscr{L}_\infty$, we can use Barbalat's lemma (Slotine and Li, 1991) to show that $\lim_{t\to\infty} r(t)=0$. Due to the form of (5), standard linear control techniques (Desoer and Vidyasagar, 1975) can be applied to show that $\lim_{t\to\infty} e(t), \dot{e}(t)=0$. □

Remark 3. A typical parameter adaptation algorithm which satisfies the conditions stated in Theorem 2 (i.e., $\hat{\theta}(t)\in\mathscr{L}_\infty$ and $\varepsilon(t)\in\mathscr{L}_2$) is the least-squares estimator given by

$$\dot{\hat{\theta}} = \frac{PY_f^{\mathrm{T}}\varepsilon}{1+\gamma\, tr\{Y_f P Y_f^{\mathrm{T}}\}}, \quad \dot{P} = -\frac{PY_f^{\mathrm{T}}Y_f P}{1+\gamma\, tr\{Y_f P Y_f^{\mathrm{T}}\}}, \tag{39}$$

where $P(t)$ is a $p\times p$ time-varying symmetric matrix, γ is a nonnegative scalar constant, and $\mathrm{tr}\{\cdot\}$ denotes the trace of a matrix. The reader is referred to Middleton and Goodwin (1988) for the proof. Note that if $\gamma=0$ in (39), we obtain the standard *unnormalized* least-squares estimator.

Remark 4. In order to satisfy the PD condition on $\hat{M}(q)$ stated in (ii) of Theorem 2, a projection-type algorithm (see Bridges et al., 1995; Lozano Leal and Brogliato, 1992 for examples) will have to be incorporated in the design of $\hat{\theta}(t)$. However, note that unlike (Middleton and Goodwin, 1988), the proposed controller does not require a projection algorithm to guarantee the boundedness of the system signals; rather, the projection algorithm is only required to illustrate asymptotic link position tracking. In addition, the control law (i.e., (12)) does not require the online calculation of $\hat{M}^{-1}(q)$.

6. Controller performance evaluation

The performance of torque input control law of (12) combined with the unnormalized least-squares estimator (i.e., (39) with $\gamma=0$) (hereinafter denoted as *LS* controller) was evaluated for the Integrated Motion, Inc. (IMI) (Direct Drive Manipulator Research and Development Package Operations Manual, 1992), 2-link, direct-drive, planar robot having the following dynamics:

$$\begin{bmatrix} p_1+2p_3c_2 & p_2+p_3c_2 \\ p_2+p_3c_2 & p_2 \end{bmatrix}\begin{bmatrix}\ddot{q}_1\\ \ddot{q}_2\end{bmatrix} + \begin{bmatrix} -p_3s_2\dot{q}_2 & -p_3s_2(\dot{q}_1+\dot{q}_2) \\ p_3s_2\dot{q}_1 & 0\end{bmatrix}\begin{bmatrix}\dot{q}_1\\ \dot{q}_2\end{bmatrix} + \begin{bmatrix} f_{d1} & 0 \\ 0 & f_{d2}\end{bmatrix}\begin{bmatrix}\dot{q}_1\\ \dot{q}_2\end{bmatrix} = \begin{bmatrix}\tau_1\\ \tau_2\end{bmatrix}, \tag{40}$$

where $p_1 = 3.473\ \mathrm{kg\,m^2}$, $p_2 = 0.193\ \mathrm{kg\,m^2}$, $p_3 = 0.242\ \mathrm{kg\,m^2}$, $f_{d1} = 5.3\ \mathrm{N\,m\,sec}$, $f_{d2} = 1.1\ \mathrm{N\,m\,sec}$, c_2 denotes $\cos(q_2)$, and s_2 denotes $\sin(q_2)$. Based on (40), the unknown parameter vector defined in (4) was arranged as $\theta = [p_1, p_2, p_3, f_{d1}, f_{d2}]^{\mathrm{T}}$. For comparison purposes, Slotine's well-known adaptive controller (Slotine and Li, 1991) which utilizes a filtered tracking error-based gradient update law was also simulated. This adaptive controller (hereinafter denoted as *GR* controller) is given by (Slotine and Li, 1991)

$$\tau = Y_s\hat{\theta} + Kr \quad \text{and} \quad \dot{\hat{\theta}} = \Gamma Y_s^{\mathrm{T}} r, \tag{41}$$

where Y_s and r were defined in (11) and (5), respectively, $K\in\Re^{n\times n}$ is a PD, diagonal, control gain matrix, and $\Gamma\in\Re^{p\times p}$ is a constant, PD diagonal, adaptation gain matrix.

In the following, both simulation and experimental results are included to illustrate the controllers performances. The inclusion of simulation results is with the objective of first clearly illustrating, without corruption from the experimental setup (e.g., noise, sensor accuracy, sampling period, etc.), the comparison between the *LS* and *GR* controllers. The experimental results are then presented with the goals of: (i) confirming the qualitative results predicted by the simulation (i.e., the intention is not to quantitatively reproduce the simulation results), and (ii) illustrating the feasibility of the real-time implementation of proposed control law.

6.1. Simulation results

The desired position trajectories for links 1 and 2 in the simulation were selected as follows:

$$q_{d1} = 1.57\sin(4t)(1-\mathrm{e}^{-0.05t^3})\ \mathrm{rad},$$
$$q_{d2} = 1.2\sin(6t)(1-\mathrm{e}^{-0.05t^3})\ \mathrm{rad} \tag{42}$$

while the initial actual link positions and velocities were set to zero. For the *LS* controller, the control gains and the initial values for the time-varying matrix $P(t)$ that

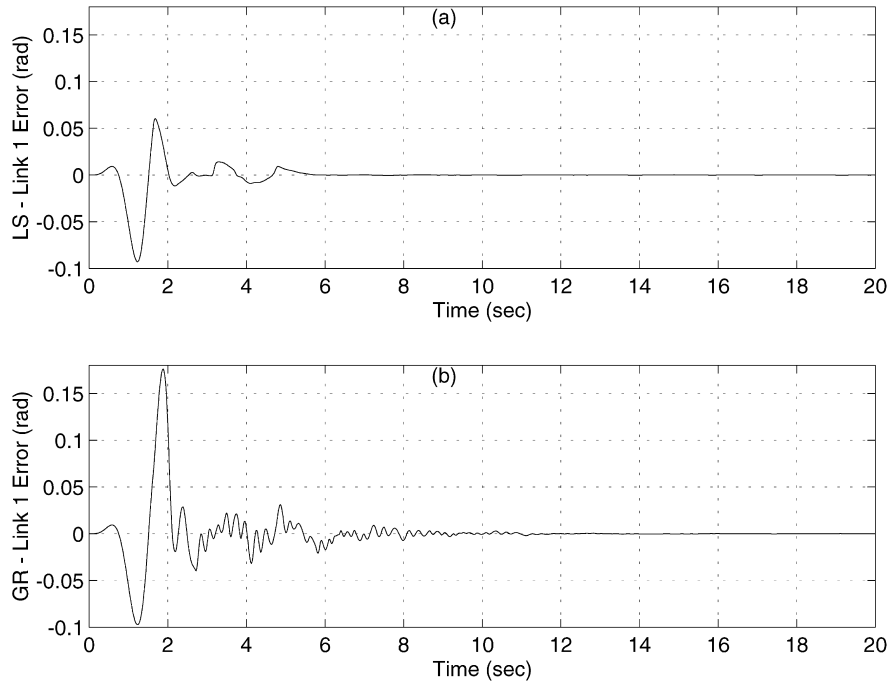

Fig. 1. e_1 (Simulation); (a) LS controller and (b) GR controller.

resulted in a good tracking performance were

$$\beta = 10, \quad \alpha = diag\{3, 5\}, \quad K = diag\{3, 3\}, \quad k_n = 0.1,$$

$$P(0) = \begin{bmatrix} 0.1 & 0.05 & 0.055 & 0.1 & 0.025 \\ 0.05 & 0.075 & 0.05 & 0.05 & 0.15 \\ 0.07 & 0.075 & 0.06 & 0.05 & 0.02 \\ 0.085 & 0.08 & 0.16 & 0.055 & 0.05 \\ 0.22 & 0.03 & 0.05 & 0.1 & 1.0 \end{bmatrix}. \quad (43)$$

For the *GR* controller, the set of control and adaptation gains which yielded good tracking performance were

$$\alpha = diag\{3, 5\}, \quad K = diag\{3, 3\},$$
$$\Gamma = diag\{2.0, 2.0, 0.8, 9.0, 3.3\}. \quad (44)$$

Note that both controllers used the same values for the feedback gains (i.e., α and K). Furthermore, for both controllers, the parameter estimate vector $\hat{\theta}(t)$ was initialized to zero. The simulation results are shown in Figs. 1–4. Figs. 1(a) and (b) illustrate the link 1 position tracking error for the *LS* and *GR* controllers, respectively, while Figs. 2(a) and (b) show the link 2 position tracking errors. Figs. 3(a) and (b) depict the parameter estimates for the *LS* and *GR* controllers, respectively. In Figs. 4(a) and (b), we have included the input control torques for links 1 and 2, respectively, of the *LS* controller while Figs. 4(c) and (d) contain the same signals for the *GR* controller. From these results, we notice the improved transient performance of the tracking error signals in the *LS* controller which can be credited to the faster and smoother parameter convergence of the least-squares update law as compared to the gradient update law.

6.2. Experimental results

The *LS* and *GR* controllers were also experimentally implemented on the IMI 2-link, direct-drive, planar robot. The desired position trajectories for links 1 and 2 were selected as

$$q_{d1} = 40.11 \sin(2t)(1 - e^{-0.3t^3}) \text{ deg},$$
$$q_{d2} = 68.75 \sin(2t)(1 - e^{-0.3t^3}) \text{ deg}. \quad (45)$$

A standard backwards difference algorithm applied to the link position measurements and then passed through a low-pass filter was used to generate the link velocity signals. The update laws given in (39) and (41) were initialized to zero while a trapezoidal algorithm was utilized to compute the integrals. The sampling period for the *LS* controller was 400 μs while the *GR* controller was run at 150 μs.[5] All control parameters were tuned by

[5] Due to its simplicity, the *GR* algorithm can be run at a faster sampling period than the *LS* algorithm. Since a faster sampling period will generally improve an algorithm's performance, this type of experimental comparison is much fairer to the *GR* algorithm.

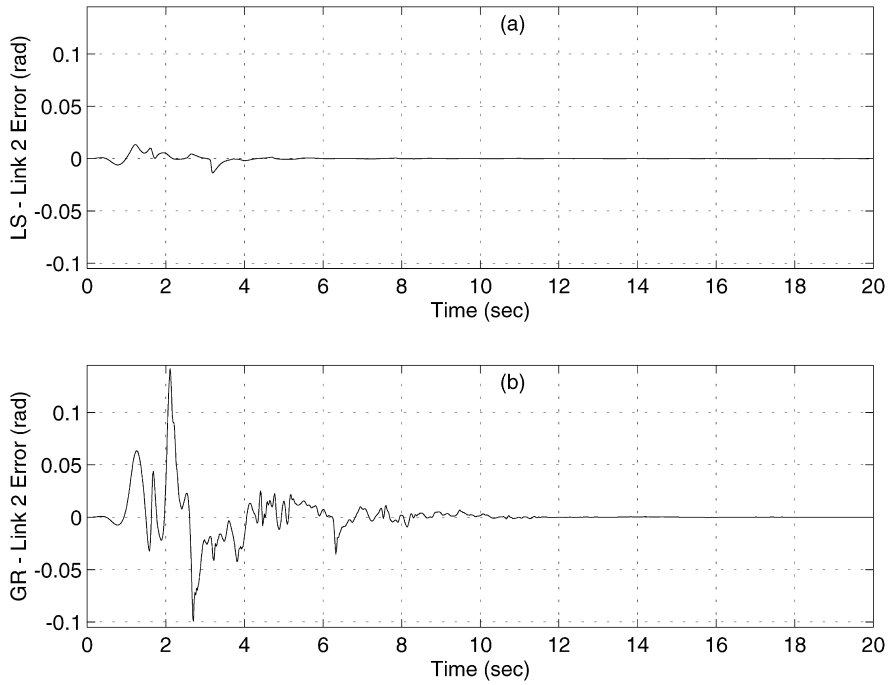

Fig. 2. e_2 (Simulation); (a) LS controller and (b) GR controller.

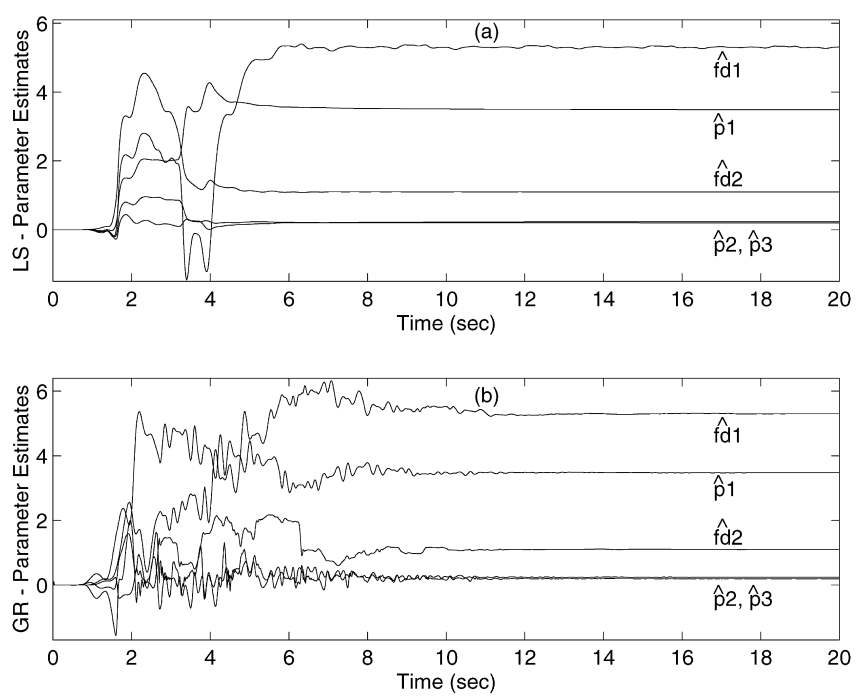

Fig. 3. $\hat{\theta}$ (Simulation); (a) LS controller and (b) GR controller.

trial-and-error until the best link position tracking performance was achieved. For the *LS* controller, the control gains and the initial values for the time-varying matrix $P(t)$ that resulted in the best tracking performance were

$$\beta = 23, \quad \alpha = diag\{51.5, 49.0\},$$
$$K = diag\{70, 26\}, \quad k_n = 1\mathrm{e}^{-4},$$

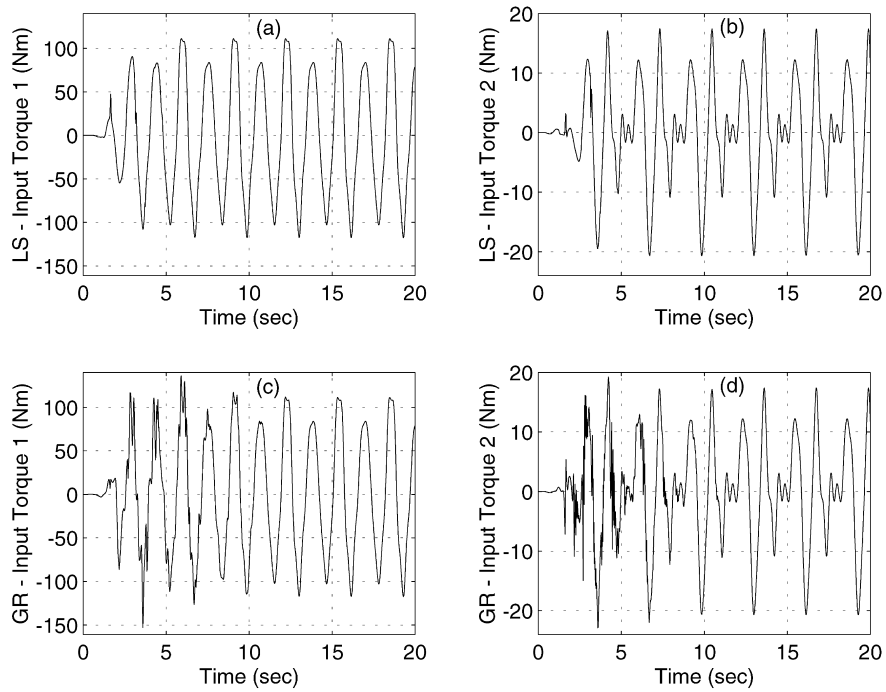

Fig. 4. τ (Simulation); (a) τ_1 - LS controller, (b) τ_2 - LS controller, (c) τ_1 - GR controller, and (d) τ_2 - GR controller.

$$P(0) = \begin{bmatrix} 0.5 & 0.25 & 0.275 & 0.5 & 0.125 \\ 0.0175 & 0.0263 & 0.0175 & 0.0175 & 0.088 \\ 0.042 & 0.045 & 0.036 & 0.03 & 0.012 \\ 0.153 & 0.144 & 0.288 & 0.099 & 0.09 \\ 0.242 & 0.033 & 0.055 & 0.01767 & 0.22 \end{bmatrix}. \tag{46}$$

For the GR controller, the set of control and adaptation gains that resulted in the best tracking performance were

$$\alpha = diag\{40, 23\}, \quad K = diag\{75, 33\}, \\ \Gamma = diag\{8.0, 0.1, 0.2, 7.0, 2.0\}. \tag{47}$$

The experimental results are shown in Figs. 5–8. Figs. 5(a) and (b) illustrate the link 1 position tracking error for the LS and GR controllers, respectively, while Figs. 6(a) and (b) show the link 2 position tracking errors. Figs. 7(a) and (b) depict the parameter estimates for the LS and GR controllers, respectively. In Figs. 8(a) and (b), we have included the input control torques for links 1 and 2, respectively, of the LS controller while Figs. 8(c) and (d) contain the same signals for the GR controller. From these figures, we again observe the faster parameter adaptation of the LS controller, and hence, the improved transient performance of the tracking error signals, as predicted by the simulation. We note that the improved tracking/parameter adaptation performance of the LS controller was obtained at the expense of only slightly larger control torques in comparison to those of the GR controller.

6.3. Discussion of results

Since the PD condition of $\hat{M}(q)$ required in part (ii) of Theorem 2 is only a *sufficient* condition for achieving asymptotic position tracking with the LS controller, it comes to no surprise that the above simulation and experimental results were obtained without utilizing a projection-type algorithm for $\hat{\theta}$, as discussed in Remark 4. Furthermore, since the requirement that $K = kI_n$ is also only a *sufficient* condition for achieving asymptotic position tracking with the LS controller, it also comes to no surprise that the experimental results were obtained without satisfying this condition (see (46)).

The experimental results indicate that an improvement in the transient response of the robot's tracking performance can be obtained through the use of the LS controller. This behavior is due in part to the indirect influence the matrix $P(0)$ has on the system's transient performance. Specifically, if higher values were utilized for the elements of $P(0)$, we observed that the parameter estimates greatly overshoot their steady-state values; thus, causing the transient tracking errors to be larger. On the other hand, if smaller values were used, the parameter estimates took longer to converge; thus, causing the tracking errors to have a larger settling time. We note that due to the 5×5 dimension and the coupled nature of $P(0)$ it is not easy to tune its elements such that a "critically damped" behavior is obtained for all the parameter estimates. However, the fact that $P(0)$ provides some means of indirectly controlling the system's transient performance in the LS controller has shown to constitute

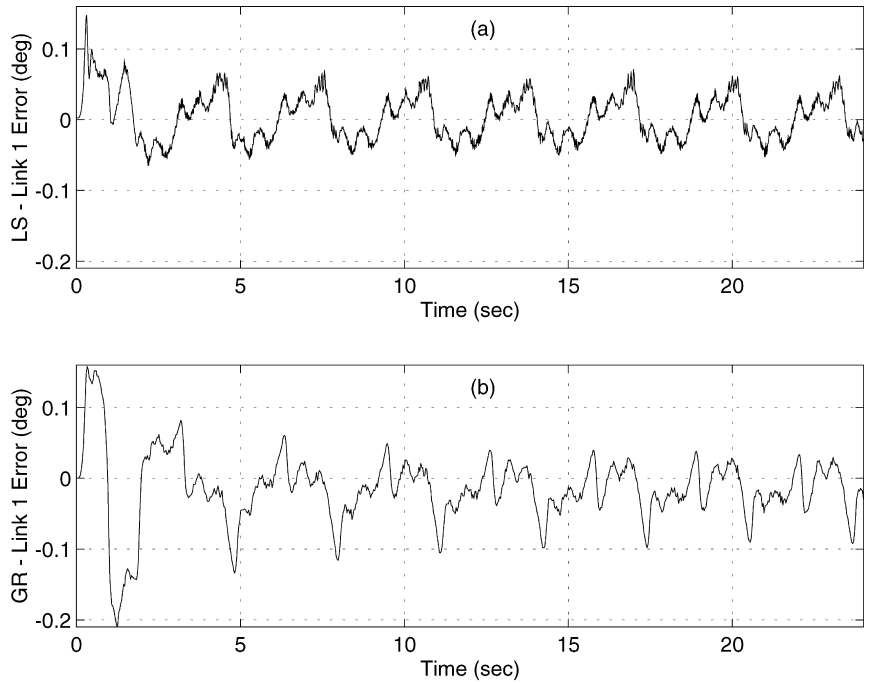

Fig. 5. e_1 (Experiment); (a) LS controller and (b) GR controller.

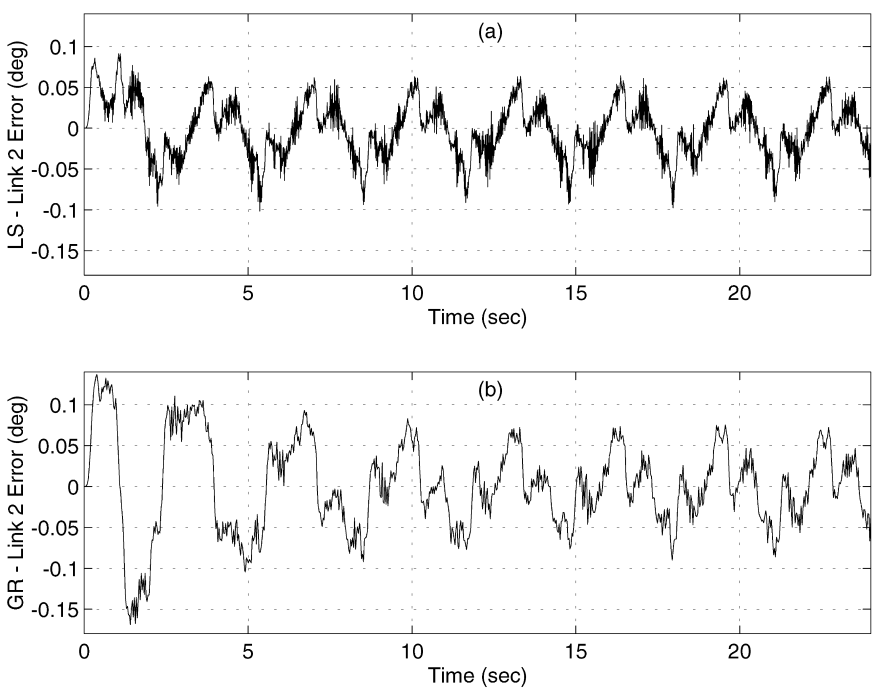

Fig. 6. e_2 (Experiment); (a) LS controller and (b) GR controller.

an advantage over the *GR* controller. As stated in the previous subsection, the control/adaptation gains and $P(0)$ were selected by trial and error to achieve the best tracking performance for the desired trajectory signal given by (45). That is, we found that the controller parameters can be tuned to obtain good link position tracking performance by monitoring the changes in the link position tracking errors and then adjusting the gains in between runs. Although the gains in (47) and (46) work for different desired trajectory signals that are reasonable (i.e., signals that limit the desired acceleration to within the capability of the motor drive system), a small amount

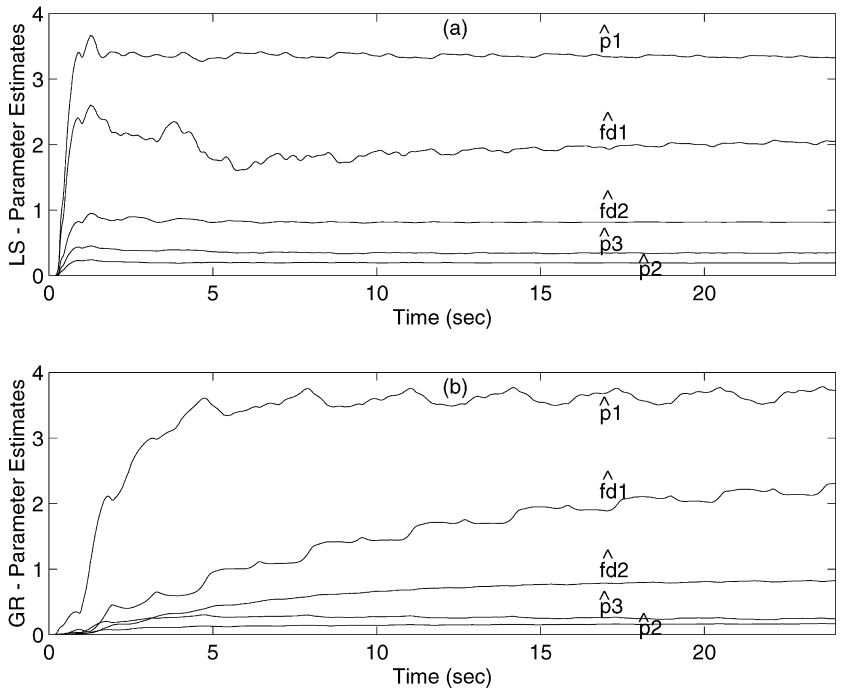

Fig. 7. $\hat{\theta}$ (Experiment); (a) LS controller and (b) GR controller.

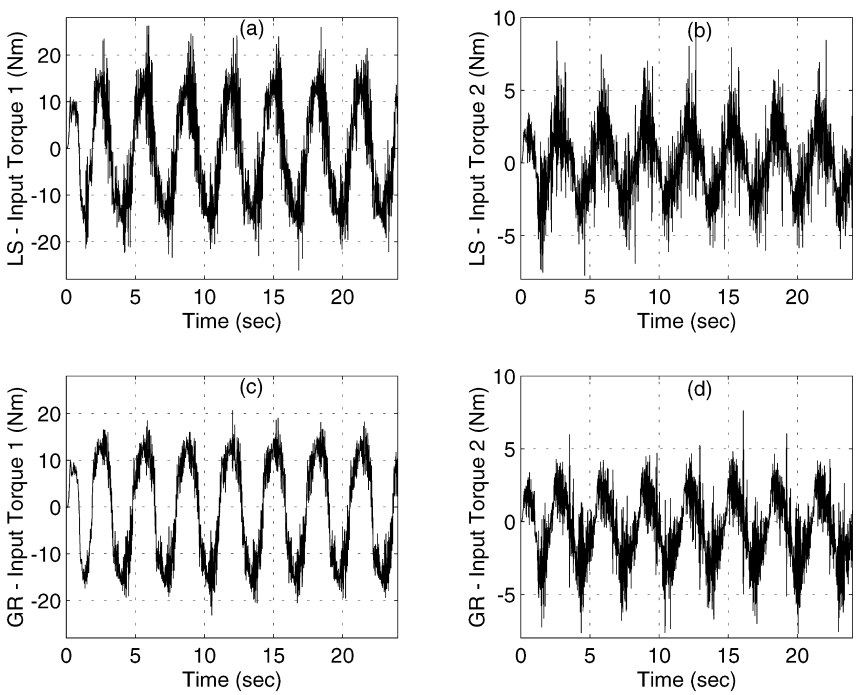

Fig. 8. τ (Experiment); (a) τ_1 - LS controller, (b) τ_2 - LS controller, (c) τ_1 - GR controller, and (d) τ_2 - GR controller.

of retuning has to be done to achieve the best link position tracking performance.

Finally, we note that the noisy behavior on the torque control inputs shown in Fig. 8 is caused by a combination of the following problems: (i) the application of a backwards difference algorithm to the link position signal for obtaining the requisite link velocity information, (ii) the use of relatively high gains to reduce the link position tracking error, and (iii) the resolution of the resolver interface board. The primary limitations on performance of the controller seem to be the resolution of the resolver, the resolver interface board, and the ability of the torque control amplifier to accurately deliver the desired torque to each robot link.

7. Conclusions

We have presented an adaptive tracking controller for robot manipulators which achieves controller/parameter update law modularity. By leveraging the work of Krstic and Kokotovic (1995), we have first shown that any prediction error-based update law which ensures the parameter estimate vector is bounded guarantees the boundedness of all closed-loop signals. Secondly, if the prediction error is square-integrable and the estimated inertia matrix is positive-definite, the proposed control law yields asymptotic link position tracking. This work represents an improvement over the design of Middleton and Goodwin (1988) since the update law does not require a projection-type algorithm to guarantee closed-loop stability; rather a projection algorithm is only required to illustrate asymptotic link position tracking. In addition, the proposed control law does not utilize the inverse of the estimated inertia matrix. The advantages of the proposed control law utilizing a standard unnormalized least-squares estimator have been illustrated through dynamic simulations and experimental results. The results show an improved tracking performance of the proposed controller in comparison to the well known, gradient update law-based, adaptive controller of Slotine and Li (1991). The improvement is due to the indirect means the least-squares estimator provides in controlling, to some extent, the system's transient performance.

Acknowledgements

This work was supported in part by the U.S. National Science Foundation Grants DMI-9457967, DDM-931133269, DMI-9622220, the Office of Naval Research Grant URI-3139-YIP01, the Square D Corporation, the Union Camp Corporation, and the AT&T Foundation.

References

Bridges, M. M., Dawson, D. M., & Abdallah, C. T. (1995). Control of rigid-link flexible-joint robots: a survey of backstepping approaches. *Journal of Robotic Systems*, *12*(3), 199–216.

Craig, J. J., Hsu, P., & Sastry, S. (1986). Adaptive control of mechanical manipulators. *IEEE conference on robotics and automation*, San Francisco, CA (pp. 190–195).

Desoer, C. A., & Vidyasagar, M. (1975). *Feedback systems: Input–output properties*. New York: Academic Press.

Direct Drive Manipulator Research and Development Package Operations Manual. (1992), Berkeley, CA: Integrated Motion Inc.

Kailath, T. (1980). *Linear systems*. Englewood Cliffs, NJ: Prentice-Hall.

Kokotovic, P. (1992). The joy of feedback: Nonlinear and adaptive. *IEEE Control Systems Magazine*, *12*, 7–17.

Krstic, M., & Kokotovic, P. V. (1995). Adaptive nonlinear design with controller-identifier separation and swapping. *IEEE Transactions on Automatic Control*, *40*(3), 426–440.

Lewis, F. L., Abdallah, C. T., & Dawson, D. M. (1993). *Control of robot manipulators*. New York: MacMillan Publishing Co.

Lozano Leal, R., & Canudas de Wit, C. (1990). Passivity based adaptive control for mechanical manipulators using LS-type estimation. *IEEE Transactions on Automatic Control*, *35*(12), 1363–1365.

Lozano Leal, R., & Brogliato, B. (1992). Adaptive control of robot manipulators with flexible joints. *IEEE Transactions on Automatic Control*, *37*(2), 174–181.

Middleton, R. H., & Goodwin, C. G. (1988). Adaptive computed torque control for rigid link manipulators. *Systems Control Letters*, *10*, 9–16.

de Queiroz, M. S., Dawson, D., & Burg, T. (1996). Reexamination of the DCAL controller for rigid link robots. *Robotica*, *14*, 41–49.

Sadegh, N., & Horowitz, R. (1990). An exponentially stable adaptive control law for robot manipulators. *IEEE Transactions on Robotics and Automation*, *6*(4), 491–496.

Slotine, J. J. E., & Li, W. (1991). *Applied nonlinear control*. Englewood Cliff, NJ: Prentice-Hall, Inc.

Tang, Y., & Arteaga, M. A. (1994). Adaptive control of robot manipulators based on passivity. *IEEE Transactions on Automatic Control*, *39*(9), 1871–1875.

Vidyasagar, M. (1978). *Nonlinear systems analysis*. Englewood Cliffs, NJ: Prentice-Hall, Inc.

Marcio S. de Queiroz was born in Rio de Janeiro, Brazil. He received a B.S. Degree in Electrical Engineering from the Federal University of Rio de Janeiro in 1990 and a M.S. Degree in Mechanical Engineering from the Pontifical Catholic University of Rio de Janeiro in 1993. In July 1997, he received a Ph.D. Degree in Electrical Engineering from Clemson University, Clemson, SC. From August 1997 to August 1998 he was a Research Associate in the Rotating Machinery and Controls Laboratory at the University of Virginia. In September 1998, he joined the Department of Mechanical Engineering at the Polytechnic University in Brooklyn, NY where he currently holds the position of visiting Assistant Professor. His research interests include nonlinear control of electromechanical systems and control of distributed parameter systems.

Darren M. Dawson was born in 1962, in Macon, Georgia. He received an Associate Degree in Mathematics from Macon Junior College in 1982 and a B.S. Degree in Electrical Engineering from the Georgia Institute of Technology in 1984. He then worked for Westinghouse as a control engineer from 1985 to 1987. In 1987, he returned to the Georgia Institute of Technology where he received the Ph.D. Degree in Electrical Engineering in March 1990. During this time, he also served as a research/teaching assistant. In July 1990, he joined the Electrical and Computer Engineering Department and the Center for Advanced Manufacturing (CAM) at Clemson University where he currently holds the position of Professor. Under the CAM director's supervision, he currently leads the Robotics and Manufacturing Automation Laboratory that is jointly operated by the Electrical and Mechanical Engineering departments. His main research interests are in the fields of nonlinear based robust, adaptive, and learning control with application to electromechanical systems including robot manipulators, motor drives, magnetic bearings, flexible cables, flexible beams, and high-speed transport systems.

Manish Agarwal received a B.Tech. Degree in Electrical Engineering from the Indian Institute of Technology, New Delhi, India in 1995 and a M.S. Degree in Electrical Engineering from Clemson University, SC in 1997. His research work at Clemson University focused on nonlinear control of flexible structures and robotic manipulators. He is currently employed as a Consultant engaged in developing web-enabled enterprise wide computing solutions in a Silicon Valley startup.